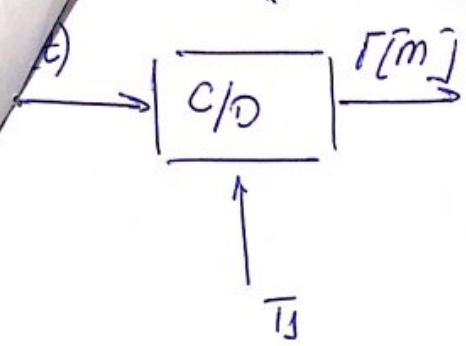
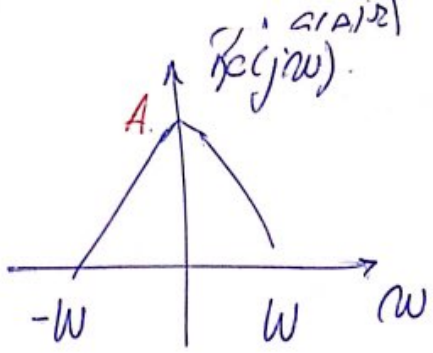
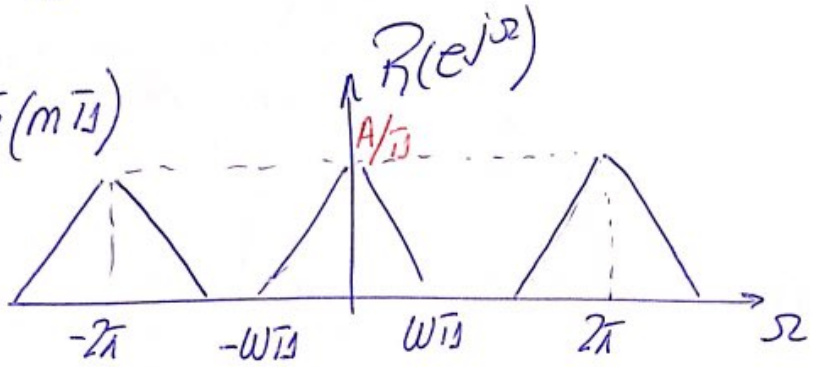




CONSIDERO QUE NO HABRÁ ALIASING.

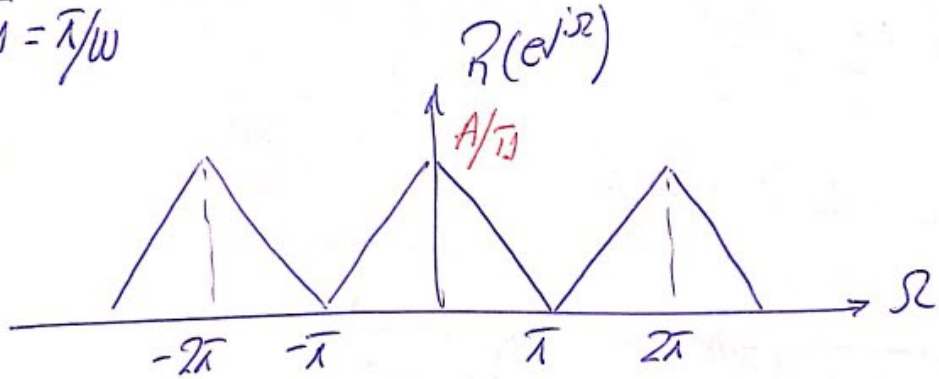
$$\Rightarrow \quad \Omega_s \geq 2\Omega \quad \Rightarrow \quad \frac{2\pi}{T_s} \geq 2\Omega \quad \Rightarrow \quad \boxed{T_s \leq \pi/\Omega}$$

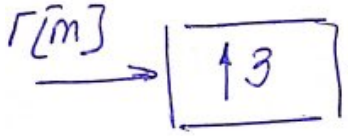
$r[m] = r_c(mT_s)$



Si $T_s = \pi/\Omega$

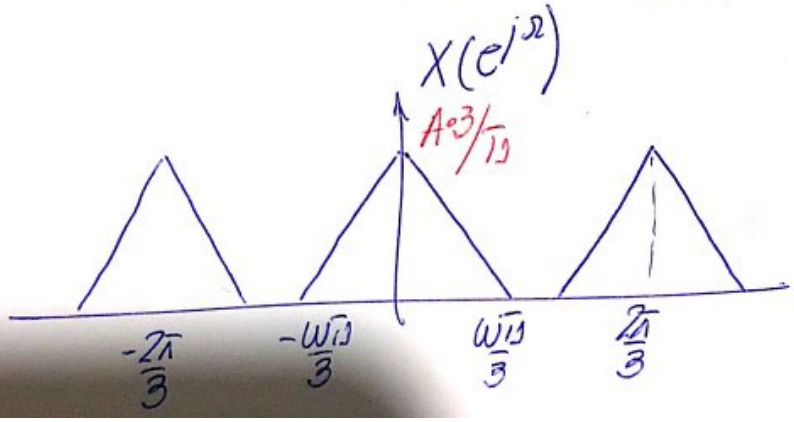
↳





$x[m] = r[m/3]$; $X(e^{j\Omega}) = X(e^{j\Omega 3})$

$x[m] = r_c\left(m\frac{T_s}{3}\right)$



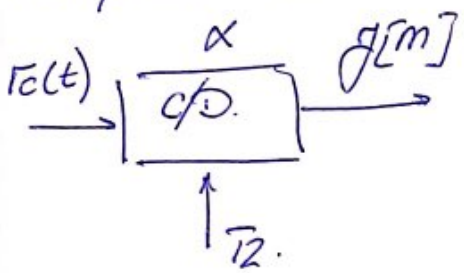
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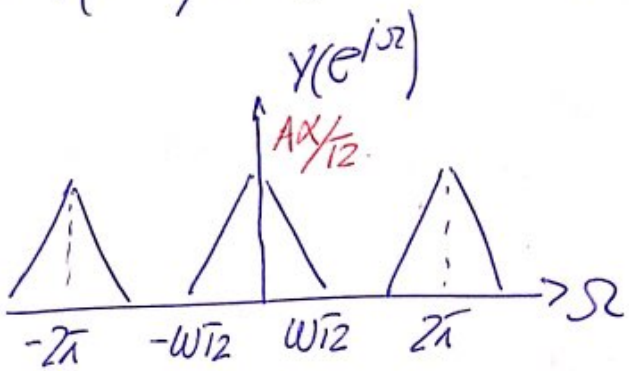
b) $g[m] = \alpha\, r_c(mT_2)$

$x[m] \rightarrow \boxed{H(e^{j\Omega})} \rightarrow g[m]$. $H(e^{j\Omega}) = \begin{cases} 1 & |\Omega| \leq \\ 0 & \text{otro.} \end{cases}$

$g[m] = x[m] * h[m] \rightarrow Y(e^{j\Omega}) = X(e^{j\Omega})\, H(e^{j\Omega})$

TENGO ENTONCES.





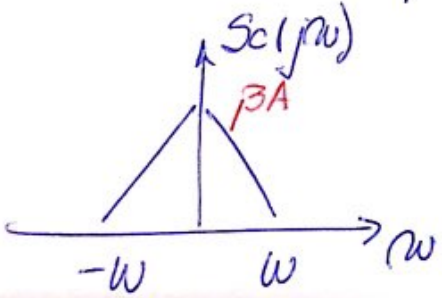
BUSCO QUE: $\frac{\omega T_1}{3} = \omega T_2 \Rightarrow \boxed{T_2 = \frac{T_1}{3}}$

$\Rightarrow \frac{A \cdot 3}{T_1} = \frac{A\alpha}{T_2} \Rightarrow \frac{A \cdot 3}{T_1} = \frac{A\alpha 3}{T_1} \Rightarrow \boxed{\alpha = 1}$

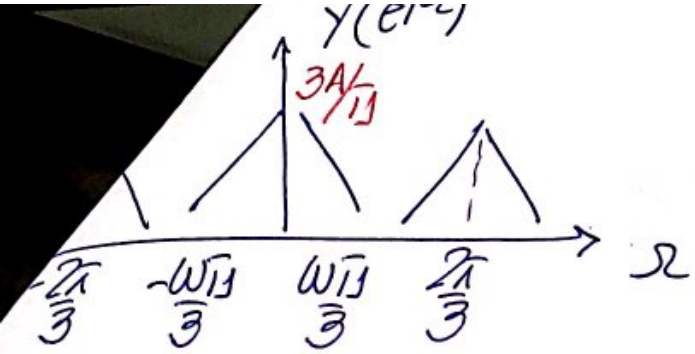
$\Rightarrow \Omega_0 = \omega T_2 = \frac{\omega T_1}{3} \Rightarrow \boxed{\Omega_0 = \frac{T_1 \omega}{3}}$

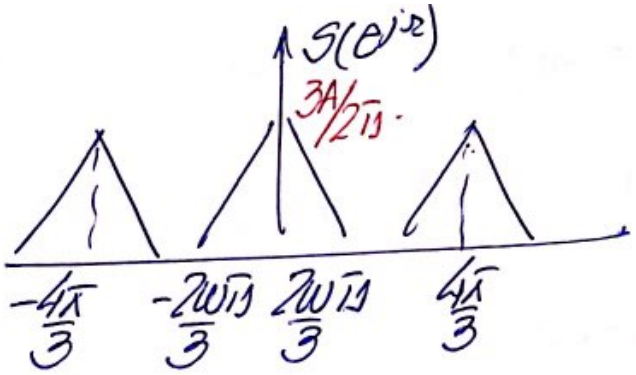
c) $g[m] \rightarrow \boxed{\downarrow 2} \rightarrow s[m] \rightarrow \boxed{\text{C/D}} \rightarrow s_c(t)$, T_3

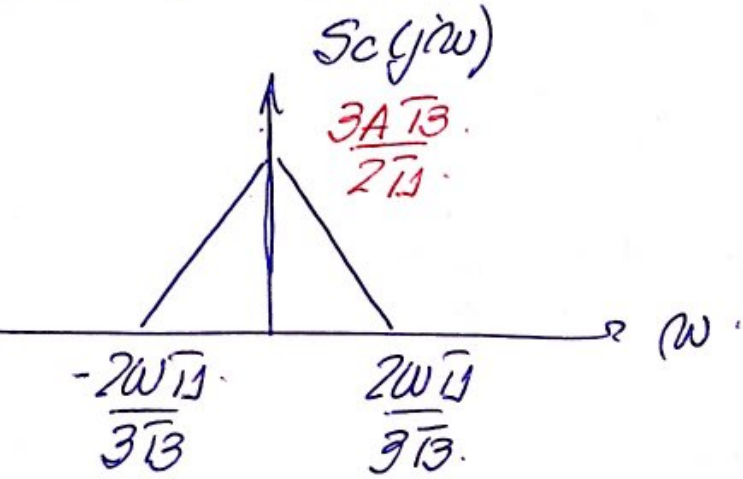
BUSCO QUE: $s_c(t) = \beta\, r_c(t) \Rightarrow S_c(j\omega) = \beta\, R_c(j\omega)$



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QUIERO QUE: • $\frac{2\omega T_3}{3T_3} = \omega \Rightarrow \boxed{T_3 = \frac{2}{3} T_1}$

• $\frac{3AT_3}{2T_1} = \beta A \Rightarrow \beta = \frac{3T_3}{2T_1} = \frac{3}{2T_1}\frac{2}{3}T_1$

↳ $\boxed{\beta = 1}$

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a) $x[n] = (-2)^n \Rightarrow y[n] = 0$

$x[n] = \left(\frac{1}{2}\right)^n u[n] \Rightarrow y[n] = \delta[n] + \alpha\left(\frac{1}{4}\right)^n u[n]$

ENTONCES:

$$x[n] = \left(\frac{1}{2}\right)^n u[n] \Rightarrow X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}$$

$$y[n] = \delta[n] + \alpha\left(\frac{1}{4}\right)^n u[n] \Rightarrow Y(z) = 1 + \frac{\alpha}{1 - \frac{1}{4}z^{-1}}$$

$$\Rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{1 + \frac{\alpha}{1 - \frac{1}{4}z^{-1}}}{\left(1 - \frac{1}{2}z^{-1}\right)^{-1}}$$

$$H(z) = \left(1 + \frac{\alpha}{1 - \frac{1}{4}z^{-1}}\right)\left(1 - \frac{1}{2}z^{-1}\right)$$

$$H(z) = \left(1 - \frac{1}{2}z^{-1}\right) + \frac{\alpha\left(1 - \frac{1}{2}z^{-1}\right)}{1 - \frac{1}{4}z^{-1}}$$

Si $x[n] = (-2)^n \Rightarrow y[n] = H(-2)\,x[n]$

DADO QUE $y[n] = 0 \Rightarrow H(-2) = 0$

$$\Rightarrow \left(1 - \frac{1}{2}\left(-\frac{1}{2}\right)\right) + \alpha\frac{\left(1 - \frac{1}{2}\left(-\frac{1}{2}\right)\right)}{1 - \frac{1}{4}\left(-\frac{1}{2}\right)} = 0$$

$$\hookrightarrow \boxed{\alpha = -9/8}$$

b) $H(1) = -1/4 \Rightarrow \boxed{y[n] = H(1)(1)^n = -1/4}$

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